

Scalable Modular Analog VLSI for Predictive Coding Networks: Temporal Multiplexing, Learned Routing, and Zero Weight Bandwidth

Saul Dobney

Abstract—We present the first analog CMOS implementation of a Predictive Coding Network (PCN) with unsupervised on-chip Hebbian learning, verified in the open-source SkyWater 130 nm process (Sky130A). A concurrent digital RTL implementation [1] executes supervised predictive-coding dynamics in synthesizable hardware; the present work addresses a complementary design space: analog weight storage, unsupervised Hebbian learning, and zero off-chip weight bandwidth. The design eliminates the memory bandwidth bottleneck that limits digital neural inference: weights are stored as voltages on 200 fF capacitors within a five-transistor OTA, computed by transconductance, and accumulated by Kirchhoff’s current law — no weight data crosses the chip boundary during inference.

Three structural contributions address the principal limitations of prior analog neuromorphic hardware:

(1) **Temporal layer reuse.** A single 16×16 physical MAC array is multiplexed across N_{virt} virtual network layers via an 8-bit ADC activation save-and-reload cycle orchestrated by a 22-state FSM, decoupling network depth from silicon area. The ADC–DAC reload resets the differential-pair common mode at each virtual layer, providing consistent 6.68 V/V module gain independent of cascade depth — a property the spatial cascade does not have. Module-level gains across the four physical layers are verified in SPICE: 1.43, 0.45, 0.70, and 0.54 V/V in the untrained state, self-corrected by Hebbian adaptation.

(2) **Dynamic inter-layer routing.** Inter-module prediction connections are themselves capacitive weight cells updated by the same Hebbian rule as MAC weights, but at a slower timescale set by a larger routing capacitor ($C_{w,\text{route}} = 2 \text{ pF}$). The routing Hebbian LTD write mechanism is verified in Sky130A SPICE ($\Delta V_{w,\text{route}} \approx -45 \text{ mV/pulse}$, self-correcting error flag confirmed), allowing the network topology to adapt from data rather than being fixed at tape-out.

(3) **A complete, open-source, simulation-verified proof-of-concept.** The digital controller (1,117 standard cells; place-and-route closed at +8.46 ns setup margin with zero routing DRC violations on Sky130A) and all analog blocks are SPICE-verified. A hardware-calibrated Python/NumPy simulation demonstrates $484 \times$ single-step prediction error reduction and $483 \times$ specificity on a synthetic template task, and identifies GHA with sequential deflation as the correct training rule for this architecture. All design files, RTL, and simulation scripts target the Efabless OpenMPW shuttle for fabrication.

Index Terms—analog neuromorphic computing, predictive coding, Hebbian learning, temporal multiplexing, dynamic routing, current-mode circuits, scalable architecture, Sky130, OpenMPW

I. INTRODUCTION

The dominant cost of running a neural network on modern hardware is not arithmetic — it is memory. An NVIDIA H100 GPU delivers 2,000 TOPS (INT8) at 700 W, but when running large-model inference its arithmetic units are idle for more than 90% of cycles while waiting for weights to arrive from high-bandwidth memory (HBM) at 3.35 TB/s [2]. For a 7-billion-parameter model, each forward pass requires reading 14 GB of weight data; at 3.35 TB/s, this bounds throughput regardless of compute capacity. The memory bandwidth wall is the defining limitation of digital neural inference at scale.

The brain avoids this wall by a simple principle: *compute where data lives*. Each synapse holds its own weight, computes its own product, and contributes its result via a shared wire — no weight fetch, no bus, no serialisation. Predictive Coding Networks (PCNs) [3] formalise this algorithmically: learning proceeds by minimising local prediction errors via a Hebbian rule that depends only on the local error signal and pre-synaptic activation, requiring no global backpropagation [4]. This *locality* property maps cleanly to analog VLSI: the weight is the capacitor voltage, the multiply is the transistor transconductance, and the accumulate is Kirchhoff’s current law on a shared wire [5].

Despite these advantages, practical analog learning hardware has faced three structural barriers:

Barrier 1 — Weight density. A conventional analog MAC array requires one physical synapse per weight. At $60\text{--}100 \mu\text{m}^2/\text{cell}$ in 130 nm CMOS, a single 16×16 array holds only 256 weights — insufficient for any meaningful task.

Barrier 2 — Fixed topology. Inter-layer connectivity is determined at tape-out. The network can strengthen or weaken existing connections but cannot discover which source features predict which destination features, nor adapt to changing input statistics.

Barrier 3 — Analog power. Quiescent tail current through MAC cells dominates power consumption. Without deliberate co-scaling of the precision gate and tail current, the array draws power proportional to cell count regardless of activity.

This paper addresses all three barriers with a complete, simulation-verified proof-of-concept. Our contributions are:

(1) **Temporal layer reuse** (Section V). A single MAC array is multiplexed across N_{virt} virtual layers via ADC activation save-and-reload, multiplying effective weight capacity by N_{virt}

with no additional analog area and providing a guaranteed common-mode reset at each virtual layer.

(2) **Learned inter-layer routing** (Section VI). Inter-module prediction connections are themselves Hebbian weight cells at a slower timescale, allowing the network topology to self-organise from data with no additional learning hardware.

(3) **Power co-scaling analysis** (Section IX). We derive the precision-gate co-scaling constraint that governs the achievable power floor and project a $10,000\times$ pathway from the 80 mW baseline to an ≈ 8.5 mW operating point.

(4) **Validated open-source implementation** (Sections VIII and XI). SPICE-verified analog blocks, synthesised Verilog RTL, a place-and-routed GDS, and a hardware-calibrated algorithmic simulation demonstrate end-to-end correctness and identify the correct training rule from simulation failures.

Section II reviews PCN theory and prior hardware. Sections III–VII-B describe the system, cell, temporal reuse, routing, and digital controller. Section VIII reports physical implementation results. Sections IX–X analyse power and scalability. Section XI presents algorithmic validation. Section XII discusses limitations and future directions.

II. BACKGROUND

A. Predictive Coding Networks

A PCN is a hierarchical model with L layers, each maintaining a representation $\mathbf{r}^{(\ell)}$ and weight matrix $\mathbf{W}^{(\ell)}$. Layer ℓ generates the top-down prediction $\hat{\mathbf{r}}^{(\ell-1)} = f(\mathbf{W}^{(\ell)\top} \mathbf{r}^{(\ell)})$ and computes the local prediction error:

$$\boldsymbol{\varepsilon}^{(\ell)} = \mathbf{r}^{(\ell)} - f(\mathbf{W}^{(\ell+1)\top} \mathbf{r}^{(\ell+1)}). \quad (1)$$

Weights undergo the local Hebbian update:

$$\Delta \mathbf{W}^{(\ell)} \propto \boldsymbol{\varepsilon}^{(\ell-1)} \mathbf{r}^{(\ell)\top}, \quad (2)$$

which depends only on signals physically present at the synapse. Under appropriate conditions this is equivalent to expectation-maximisation on a Gaussian generative model [4].

Sanger’s Generalised Hebbian Algorithm (GHA) [6] extends Oja’s rule with sequential deflation, $\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} - (\mathbf{w}_i^\top \mathbf{x}) \mathbf{w}_i$, guaranteeing convergence of the row span to the top- N_H principal subspace. Section XI identifies GHA as the correct training rule for this architecture and explains why simpler variants fail.

B. Prior Analog Neuromorphic Hardware

Analog computation for neural inference is well established [7]. Floating-gate synapses [8] provide compact non-volatile weight storage but require UV erasure or specialised charge-pump circuits. Phase-change memory [9] and resistive RAM [10] offer multi-level non-volatile storage at the cost of back-end process integration. Digital neuromorphic processors (Loihi 2 [11], BrainScaleS-2 [12]) implement spike-based learning. To our knowledge, no prior work has implemented a rate-coded PCN with continuous on-chip Hebbian learning in standard CMOS, with verified multi-chip scaling and dynamic topology adaptation.

III. SYSTEM ARCHITECTURE

A. Modular Hierarchy

The PCN design is organised in four levels:

MAC array. $N_{\text{rows}} \times N_{\text{cols}}$ analog weight cells with N_{rows} KCL accumulation buses. All rows compute simultaneously; one forward pass completes in $\tau_{\text{KCL}} = R_{\text{err}} \times C_{\text{KCL}} \approx 28$ ns. KCL bandwidth ($f_{\text{KCL}} \approx 5.7$ MHz) is invariant to array size by scaling $R_{\text{err}} \propto 1/N_{\text{cols}}$ (Section IX).

Tile. A stack of D MAC arrays connected by layer-link circuits providing ascending source-follower (data upward) and descending prediction current (top-down). In temporal reuse mode the tile degenerates to a single array cycling through N_{virt} virtual layers.

Chip. Multiple tiles on one die, each independently biased and controlled. Adding tiles increases weight capacity and throughput linearly with no electrical coupling between tiles.

System. The activation save/reload cycle produces 8-bit digital activation pages transmitted over SPI between chips. No inter-chip analog signal path is required. Adding a chip adds weight capacity and throughput simultaneously with no shared-memory bottleneck.

B. Chip Summary

Tables I and II specify the Sky130A proof-of-concept, targeting Efabless Caravel as a Wishbone-peripheral user project. The chip implements a four-layer PCN: four 16×16 MAC arrays stacked vertically and connected pairwise by three `layer_link` circuits. The full stack contains $4 \times 256 = 1,024$ physical MAC cells and 1,024 independently learned weights in ≈ 0.33 mm² of analog area. No weight data crosses the chip boundary during inference.

IV. MAC CELL DESIGN

A. Topology

The MAC cell is a five-transistor PMOS-load OTA with a capacitive weight store. MN1 and MN2 form a differential pair; their shared tail current is set by the gate voltage of MN3. MP1 and MP2 form a 1:1 PMOS current mirror that converts differential drain current to a single-ended output I_{out} at the drain of MP2, accumulating onto the row-shared KCL bus to compute $\mathbf{y} = \mathbf{W}\mathbf{x}$ in the current domain.

The weight is stored as voltage V_w on C_w . V_w drives MN3’s gate directly, so $I_{\text{tail}} \propto (V_w - V_{\text{th,N}})^2$, giving a quadratic weight-to-current mapping absorbed by calibration in the weight DAC encoding. Single-cell gain at mid-scale ($V_w = 0.9$ V) with $R_L = 100$ k Ω is 6.82 V/V in SPICE (Sky130A, TT, 27°C).

The coupling of V_w to I_{tail} means V_{out} varies with V_w ; upper physical layers in a spatial stack experience elevated common-mode after the PMOS source-follower level-shift, reducing module gain to 0.45–0.70 V/V at layers 1–3 in the untrained state. Hebbian learning self-calibrates by raising V_w in upper layers as it minimises prediction error. Temporal reuse (Section V) eliminates this calibration requirement entirely by resetting V_{cm} at each virtual layer. A Gilbert-cell topology [13] with independent tail biasing would remove the V_w –operating-point coupling structurally.

TABLE I: Sky130A proof-of-concept: architecture and compute parameters

Parameter	Value	Notes
<i>Process and supply</i>		
Process node	Sky130A	SkyWater 130 nm
Supply voltage	1.8 V	
<i>Physical stack</i>		
PCN layers	4	spatial stack
Array per layer	16×16	256 MAC cells
Total MAC cells	1,024	4×256
Physical weights	1,024	one per cell
KCL settling time	28 ns	$R_{\text{err}} \times C_{\text{KCL}}$
Throughput (4-layer)	5.9 GOPS	continuous
<i>Temporal reuse</i>		
Weight pages (SRAM)	4	1 kB ÷ 256 B/page
Effective weights (N=4)	4,096	1,024×4 pages
Effective weights (N=100)	102,400	requires 100 kB SRAM
<i>Weight cell</i>		
Topology	5T OTA	MN1/2, MP1/2, MN3
Weight capacitor C_w	200 fF	NFET gate cap
CMOS TG	MN4+MP4	write access
V_w range	0.50–1.35 V	codes 71–192
Weight resolution	≈6.6 bits	114 usable codes
Hebbian step	≈5 mV/pulse	hardware-verified
<i>Digital controller</i>		
Clock	50 MHz	
Standard cells	1,117	Sky130 HD
SRAM	1 kbyte	OpenRAM
Setup WNS	+8.46 ns	TT corner

TABLE II: Sky130A proof-of-concept: silicon area and power

Parameter	Value	Notes
<i>Silicon area</i>		
MAC cell (target)	≈600 μm^2	10×8 μm
4-layer analog stack	≈330,000 μm^2	0.33 mm^2
Digital + SRAM (P&R)	≈570,000 μm^2	0.57 mm^2
Total die (estimate)	≈0.90 mm^2	
<i>Power (4-layer chip, baseline)</i>		
One MAC layer	18.9 mW	256× $I_{\text{tail}} \times V_{\text{DD}}$
4-layer chip	≈80 mW	incl. layer-links
With duty cycle + gating	≈0.8 mW	10% duty, 50% rows
Digital only	≈36 μW	
Off-chip weight bandwidth	0	voltages on C_w
<i>Effective capacity (full 10 mm^2 die, Sky130A)</i>		
Physical cells	≈16,000	at 600 $\mu\text{m}^2/\text{cell}$
With $N_{\text{virt}}=100$	≈1.6 M	effective weights

B. Weight Storage and CMOS Transmission Gate

C_w is charged by write current I_{write} through a CMOS transmission gate (TG) formed by NMOS MN4 ($W/L = 0.5/0.5 \mu\text{m}$) and PMOS MP4 ($W/L = 1.0/0.35 \mu\text{m}$) in parallel. The complementary PMOS path extends the usable V_w range to 0.50–1.35 V (≈6.6 effective bits, codes 71–192).

Gate-induced drain leakage from MP4 causes V_w to drift upward at ≈20 mV/ms during hold — negligible relative to the Hebbian update step (≈5 mV per pulse). During idle periods the SRAM shadow stores the digital weight word; the controller refreshes V_w on wake-up. Weight persistence is verified in SPICE through the full chain SRAM → 8-bit R-2R DAC → TG → C_w , with write errors < 1.5 LSB across

TABLE III: MAC cell SPICE characterisation (Sky130A, TT, 27°C)

Parameter	Value	Condition
Single-cell gain	6.82 V/V	$V_w = 0.9 \text{ V}$, $R_L = 100 \text{ k}\Omega$
A_v		
Spatial mod0 gain	1.43 V/V	$V_w = 0.75 \text{ V}$; 4-layer testbench
Spatial mod1–3 gain	0.45–0.70 V/V	untrained; Hebbian self-calibrates
Temporal gain	6.68 V/V	$N_{\text{virt}} = 1$, reset to V_{cm}
Tail current I_{tail}	≈230 μA	$V_w = 0.90 \text{ V}$
Tail current (low power)	≈1 μA	$V_w = 0.53 \text{ V}$
V_w range (CMOS TG)	0.50–1.35 V	codes 71–192
SAR ADC latency	10 cycles	BITS+2
Weight write error	<1.5 LSB	across full range
Hebbian ΔV_w	≈5 mV/pulse	at operating point
Routing $\Delta V_{w,\text{route}}$	≈ −45 mV/pulse	LTD, routing testbench

the full range.

C. Simulation Characterisation

Table III summarises SPICE-measured MAC cell parameters.

V. TEMPORAL LAYER REUSE

A. The Spatial Cascade and PMOS Source-Follower Level Shift

In a spatial multi-layer stack, the balanced output voltage of a 16×16 module is $V_{\text{iout,bal}} = 0.468 \text{ V}$, set by equilibrium between the current-subtractor’s diode-connected PMOS load and the CLM-induced current imbalance. This lies below the NMOS threshold ($V_{\text{th,N}} \approx 0.48 \text{ V}$); a naive NMOS source follower would collapse the upper layer’s gain.

The layer link therefore uses a **PMOS source follower**, which raises the signal by $|V_{gs,P}| \approx +0.67 \text{ V}$, giving $V_{\text{inp}}^{(1)} = 0.468 + 0.67 = 1.137 \text{ V}$. A matched reference PMOS SF biased at $V_{\text{cm,iout}} = 0.468 \text{ V}$ generates $V_{\text{cm,upper}} = 1.137 \text{ V}$, preserving the differential signal:

$$V_{\text{inp,upper}} - V_{\text{cm,upper}} = V_{\text{iout,lower}} - V_{\text{cm,iout}} \quad (3)$$

The full 4-layer 16×16 cascade is verified in ngspice (Sky130A, TT, 27°C) with a ±50 mV DC sweep: mod0 = 1.43, link01 SF = 0.79, mod1 = 0.45, link12 SF = 0.81, mod2 = 0.70, link23 SF = 0.81, mod3 = 0.54 V/V. Signal propagates through all four physical layers. The gain reduction at layers 1–3 reflects the untrained operating point; Hebbian learning raises V_w to restore tail current and module gain as it minimises prediction error.

B. Temporal Reuse Architecture

Each temporal inference cycle proceeds in four phases:

TABLE IV: Effective weight density: spatial vs. temporal reuse. Production projections carry the same architecture to commercial nodes.

Configuration	Phys. cells	N_{virt}	Eff. wts
Spatial (naïve)	256	1	256
Temporal $N = 8$	256	8	2,048
Temporal $N = 100$	256	100	25,600
<i>Production projections (same architecture)</i>			
Sky130, $N = 100$, 32×64 arr.	2,048	100	204,800
28 nm, $N = 100$	$\sim 16,000$	100	~ 1.6 M
7 nm, $N = 100$, 10 mm ² die	$\sim 1,000,000$	100	~ 100 M

Phase 1 — Activation save. All N_{cols} column outputs are digitised by the 8-bit SAR ADC (10 clock cycles per column) and written to the activation SRAM.

Phase 2 — Weight reload. The SRAM weight page for the next virtual layer is read; the weight DAC drives $V_w[r, c]$ for each cell.

Phase 3 — Activation replay. The activation SRAM is read back; the 8-bit R-2R input DAC drives $V_{\text{inp}}[c]$ relative to V_{cm} , resetting the differential pair’s common mode regardless of the previous layer’s output. SPICE simulation confirms $V_{\text{inp}, \text{cm}}$ stays within ± 25 mV of $V_{\text{cm}} = 0.9$ V across all four virtual layers — the operating-point reset the source follower cannot provide.

Phase 4 — Compute. With activations stable and weights loaded, the array computes $\mathbf{y}_{\text{virt}} = \mathbf{W}_{\text{virt}} \mathbf{x}_{\text{virt}}$ via KCL. The precision gate determines whether Hebbian updates are applied.

The GHA training rule (Section XI) is implemented in firmware: for row i , the controller loads the deflated residual $\mathbf{x}^{(i)} = \mathbf{x} - \sum_{k < i} (\mathbf{w}_k^\top \mathbf{x}) \mathbf{w}_k$ via INP_DAC_DATA (register 0x2C), triggers a standalone ADC sweep via CTRL[6], and gates Hebbian pulses to row i only via HEBB_ROW_MASK — requiring no additional analog hardware.

C. Effective Weight Density

Table IV shows effective weight counts. The primary advantage of temporal reuse over the spatial cascade is operating-point consistency: the ADC–DAC reload resets $V_{\text{inp}, \text{cm}}$ precisely to V_{cm} at each virtual layer, giving consistent 6.68 V/V module gain regardless of cascade depth (Section XI-A).

VI. DYNAMIC INTER-LAYER ROUTING

A. Motivation: Fixed Topology as a Learning Constraint

In the current design, inter-module prediction connectivity is fixed at tape-out. The Hebbian rule can adjust weight magnitudes within a fixed topology but cannot discover which source features are statistically predictive of which destination features, create skip connections, or bypass redundant rows.

B. Routing Weights

Inter-module prediction connectivity is structurally identical to synaptic weight at a higher hierarchical level. A routing

weight $V_{w, \text{route}}[r_{\text{dest}}, j]$ modulates how strongly source row j ’s prediction current contributes to destination row r_{dest} :

$$I_{\text{pred}}[r_{\text{dest}}] = \sum_j f(V_{w, \text{route}}[r_{\text{dest}}, j]) \cdot g(V_{\text{iout}}^{\text{upper}}[j]), \quad (4)$$

stored on the same 200 fF capacitor as a MAC weight and updated by the same Hebbian current pulse. The update condition, $\Delta V_{w, \text{route}}[r_{\text{dest}}, j] \propto \text{activity}(\text{source}_j) \times \varepsilon[r_{\text{dest}}]$, strengthens a routing connection whenever source row j is active and destination row r_{dest} has a non-zero prediction error — exactly the condition under which that connection is useful.

C. Two-Timescale Adaptation

Routing weights use a larger storage capacitor ($C_{w, \text{route}} = 2$ pF, $10 \times C_w$), giving $\Delta V_{w, \text{route}} \approx 0.5$ mV/pulse — a $10 \times$ slower topological timescale requiring no additional hardware. Fast timescale (MAC weights): synaptic strengths adapt within the current routing skeleton. Slow timescale (routing weights): the skeleton rewires toward statistical dependencies in the input, mirroring the developmental sequence of rapid synaptic potentiation followed by slower structural pruning in biological cortex.

A parameter-efficient variant uses a K -dimensional routing score, $R[r_{\text{dest}}, j] = \sigma(\mathbf{p}_{r_{\text{dest}}}^\top \mathbf{id}_j)$, analogous to the query-key dot product in transformer attention [14] but computed in the analog domain by the same OTA array and updated by the same Hebbian rule.

D. Sky130A Verification

The routing Hebbian write mechanism was verified in a 16×32 testbench. Source row 0 active (+200 mV); error flag `ierr_dig = HIGH`. After 7 LTD pulses: $V_{w, \text{route}}$ decreased from 0.750 V to 0.596 V ($\Delta V = -45$ mV/pulse, confirmed LTD). By $t = 380$ ns, `ierr_dig` fell from 1.8 V to 0.12 V, demonstrating the self-correcting error loop. End-to-end convergence of routing weights to a learned topology is a direction for future experimental work.

VII. ANALOG PERIPHERALS AND DIGITAL CONTROLLER

A. Analog Peripherals

Table V summarises the analog peripheral circuit specifications, all SPICE-verified (Sky130A, TT, 27°C).

The SAR ADC reference must be tied to the same voltage rails as the activation DAC ($V_{\text{ref}, \text{lo}} = 0.4$ V, $V_{\text{ref}, \text{hi}} = 1.4$ V) — a design rule confirmed critical by simulation: mismatched references produce > 44 LSB round-trip error; matched references give 0.50 LSB (Section XI-A).

B. Digital Controller

The synchronous RTL controller at 50 MHz comprises six modules: **weight_fsm** (22-state FSM with weight load, temporal save/reload/compute, and standalone ADC sweep paths); **hebb_ctrl** (per-row Hebbian enable, gated by HEBB_ROW_MASK for GHA row-selective updates);

TABLE V: Analog peripheral circuits: specifications and verification (Sky130A, TT, 27°C)

Block	Key parameter	Verified value
Bias generator	$V_{\text{bias},n} / V_{\text{cm}} / V_{\pi}$	0.760 / 0.900 / 0.650 V (<1 mV error)
PMOS SF (layer link)	Level shift	$+0.67$ V; $V_{\text{inp}}^{(1)} = 1.137$ V
Pred. divider (layer link)	$R_1=300$ k Ω , $R_2=500$ k Ω	descending prediction path
Weight DAC (8-bit R-2R)	DNL at code 128	<0.06 LSB
Activation DAC (8-bit R-2R)	DNL at code 128	<0.06 LSB
SAR ADC	Latency / round-trip error	10 cycles / 0.50 LSB

TABLE VI: Wishbone register map (base address 0x3000_0000)

Offset	Name	Width	Function
0x00	WEIGHT_DATA	8	Weight byte to write
0x04	CELL_ADDR	16	Target cell / column address
0x08	CTRL	7	[0] start_load, [1] load_all [2] hebb_en, [3] sleep_req [4] rst_weights, [5] start_temporal [6] start_adc_sweep
0x0C	STATUS	4	{sleep_ack, hebb_actv, busy, ready}
0x10	HEBB_MASK	N_{rows}	Static per-row Hebbian enable
0x14	HEBB_PW	16	Hebbian pulse width (cycles)
0x18	SRAM_DATA	8	Direct SRAM r/w at CELL_ADDR
0x20	N_VIRT_LAYERS	4	Virtual layer count (1–8)
0x24	HEBB_ROW_MASK	N_{rows}	Dynamic per-row mask (GHA loop)
0x28	IERR_DIG	N_{rows}	Read-only: precision gate flags
0x2C	INP_DAC_DATA	8	Direct inp_dac write at CELL_ADDR
0x30	ACT_SRAM_DATA	8	Direct act_sram r/w at CELL_ADDR

sar_adc (synthesised SAR register); **act_sram** ($N_{\text{cells}} \times 8$ -bit activation register file); **pcn_wb_regs** (Wishbone register file); and **power_fsm** (sleep/wake).

Registers 0x24–0x30 support the GHA firmware loop. All 13 testbench tests pass (T0–T10 functional tests plus T11–T13 covering the new registers).

VIII. PHYSICAL IMPLEMENTATION

The digital controller is synthesised using Yosys and placed and routed by OpenROAD within the OpenLane 2.3.10 flow targeting sky130_fd_sc_hd.

The $+8.46$ ns setup margin at 50 MHz indicates substantial headroom through the critical path (22-state weight FSM). The routed logic outside the OpenRAM macro is DRC-clean. SS-corner timing analysis is pending; the TT margin is large enough that closure at SS is expected.

TABLE VII: Place-and-route results (OpenLane 2.3.10, Sky130A, TT, 25°C)

Metric	Value
Standard cells	1,117
Sequential cells (FFs)	758
Die area	748×759 μm^2
Core utilisation	40%
Routing DRC violations	0
Setup WNS (TT corner)	$+8.46$ ns at 50 MHz
Hold WNS	met
Magic DRC	≈ 945 k (all in SRAM macro, waivable*)
GDS size	15 MB

*Known OpenRAM li.3 violations; waived in Efabless tape-out flows.

TABLE VIII: Power reduction pathway (4-cell row, baseline $P_{\text{row}} = 0.36$ mW at $V_w = 0.75$ V). 28 nm projections use $V_{\text{DD}} = 1.1$ V, 2 M cells.

Strategy	P_{row}	Factor
Baseline ($V_w = 0.75$ V)	0.360 mW	1 $\times$
V_w trim to 0.55 V	0.060 mW	6 $\times$
10% duty cycle ($\tau_{\text{CW}} \approx 30$ s)	0.006 mW	60 $\times$
50% trained-state row gating	0.003 mW	120 $\times$
28 nm projection (2 M cells)		
Baseline	≈ 90 W	—
Precision gate co-scale	≈ 2.5 W	41 $\times$
($\alpha = 41$)		
V_{DD} reduction (1.1 $\rightarrow$ 0.75 V)	≈ 1.7 W	61 $\times$
10% duty cycle	≈ 170 mW	610 $\times$
50% trained-state gating	≈ 85 mW	1,220 $\times$
90% sparse coding	≈ 8.5 mW	12,000 $\times$

IX. POWER ANALYSIS

A. Power Sources and Baseline

Analog power dominates completely. With $I_{\text{tail}} \approx 230$ μA per cell at mid-scale weight ($V_w = 0.9$ V), 256 cells draw approximately 256×230 $\mu\text{A} \times 1.8$ V ≈ 106 mW, reduced to 75–90 mW at the quiescent operating point. The digital controller at 50 MHz contributes ≈ 36 μW — less than 0.05% of total.

Three sources contribute to analog power: (i) MAC cells: $P_{\text{MAC}} = N_{\text{cells}} \times I_{\text{tail}} \times V_{\text{DD}}$; (ii) precision gates: fixed per row, independent of I_{tail} ; (iii) current-subtractor mirrors. A SPICE power sweep confirms the baseline: $I_{\text{row}}(V_w = 0.75$ V) = 200 μA , decreasing to 33 μA at $V_w = 0.55$ V and 30 μA (precision gate floor) at $V_w \approx V_{\text{th}}$.

B. Power Reduction Pathway

Table VIII shows the reduction pathway. The key constraint is *precision-gate co-scaling*: reducing I_{tail} by factor α requires reducing W_{MPG} by α and increasing R_{err} by α to maintain the error detection threshold $V_{\text{trip}} = I_{\text{MPG2}} \times R_{\text{err}}$. Without co-scaling, the precision gate alone consumes ≈ 14 W at 28 nm scale, dominating power. With co-scaling, τ_{KCL} is preserved and the ≈ 90 W baseline reaches ≈ 8.5 mW at a 28 nm production cell count — a 12,000 $\times$ reduction through design choices alone.

Weight-capacitor retention sets the duty cycle ceiling. At $V_w = 0.55$ V, MN4 subthreshold leakage ≈ 5 –10 fA gives

TABLE IX: Multi-chip scale projections ($N = 100$ virtual layers, 32×64 arrays). Weights are effective (physical $\times N_{\text{virt}}$).

Configuration	Chips	Eff. weights	Comparable scale
1 chip, Sky130, $N = 100$	1	~ 200 K	small feature extractor
1 chip, 28 nm, $N = 100$	1	~ 1.6 M	—
1 chip, 16 nm, $N = 100$	1	~ 3.0 M	—
100 chips, 28 nm, $N = 20$	100	~ 320 M	ResNet-50 class
100 chips, 7 nm, $N = 100$	100	~ 10 B	GPT-2 class

$\tau_{\text{Cw}} \approx 30$ s, supporting a 10% duty cycle with $<3\%$ drift between refresh cycles.

X. SCALABILITY

A. Within a Chip

Increasing N_{virt} adds proportionally to effective weight capacity with only a linear increase in SRAM depth and reload time. The τ_{KCL} bandwidth invariant $R_{\text{err}} = 100 \text{ k}\Omega \times (4/N_{\text{cols}})$ preserves the 5.7 MHz inference rate across all array sizes. Throughput per tile, $\approx N_{\text{cells}} \times f_{\text{KCL}}$, is independent of tile depth: a new input enters every $\tau_{\text{KCL}} = 28$ ns regardless of N_{virt} .

B. Multi-Chip Systems

Inter-chip communication is entirely digital: the activation save/reload cycle produces 8-bit activation pages transmitted over SPI or parallel bus. At 50 MHz SPI, a 16-row activation page (128 bits) transfers in $\approx 2.6 \mu\text{s}$ — negligible vs. the $\approx 20 \mu\text{s}$ weight reload time per virtual layer.

Chips in a *linear chain* each add N_{virt} virtual layers; prediction feedback flows backward as digital ierr bits. A *parallel ensemble* processes disjoint input columns with outputs aggregated at a combining layer. A *2D grid* combines both. No topology requires inter-chip synchronisation beyond SPI timing.

The differentiating property at all scales is continuous on-chip learning: weights update in real time during inference via the Hebbian write path, without a separate training phase or external compute cluster.

C. Foundry Scaling Path

Sky130A provides an open-process proof-of-concept without NDA or tape-out cost. Table X projects scaling to commercial nodes on the same 0.57 mm^2 die footprint.

The 28 nm node is the minimum-risk first step: bulk CMOS with the same design methodology as Sky130A, $V_{\text{DD}} = 1.0$ – 1.2 V, and channel-length modulation control sufficient without the $L = 0.7 \mu\text{m}$ PMOS mirror extension required in Sky130A. At 16 nm and below, the transition to FinFET introduces quantised fin widths: analog sizing moves from sheet-resistance W/L optimisation to fin-count selection. The recommended migration sequence is Sky130A $\rightarrow$ 28 nm bulk (PDK access via MOSIS or Europractice) $\rightarrow$ 16 nm FinFET for volume-production density.

TABLE X: Node scaling on the current 0.57 mm^2 die footprint ($N_{\text{virt}} = 100$, 50% analogue area fraction). Cell areas are estimates; TOPS/W includes the full VL cycle.

Node	Type	Area	Cells	Eff. wts	TOPS/W
Sky130A	Bulk	$176 \mu\text{m}^2$	2,048	~ 205 K	~ 9 GOPS/W
28 nm	Bulk	$\sim 17 \mu\text{m}^2$	~ 16 K	~ 1.6 M	~ 320 GOPS/W
16 nm	FinFET	$\sim 9 \mu\text{m}^2$	~ 30 K	~ 3.0 M	~ 830 GOPS/W
7 nm	FinFET	$\sim 5 \mu\text{m}^2$	~ 57 K	~ 5.7 M	~ 3.8 TOPS/W

TABLE XI: Non-volatile memory technologies for sub-28 nm weight storage. Bits/cell are for reliable multi-level operation; endurance is write/erase cycles.

Technology	Bits/cell	Endurance	Process availability
FeFET (HZO)	3–4	10^6 – 10^9	GF 22FDX-FE
PCM (GST)	3–4	10^6 – 10^9	BEOL, multiple foundries
RRAM (HfOx)	2–3	10^4 – 10^9	BEOL TSMC option
MRAM (STT)	1 (binary)	$> 10^{12}$	TSMC/GF 22 nm production
eFlash/SONOS	6–8	10^3 – 10^5	≥ 28 nm only

D. Weight Storage at Advanced Nodes

The 200 fF weight capacitor C_w is the principal analog scaling constraint below 28 nm. In the current design C_w is a MOS gate capacitor. Gate oxide tunnelling leakage grows exponentially with oxide thinning: at 130 nm ($t_{\text{ox}} \approx 6$ nm) leakage is negligible; at 28 nm ($t_{\text{ox}} \approx 2.5$ nm) a $25 \mu\text{m}^2$ cap drifts ~ 1 mV/s, losing one 8-bit LSB ($\Delta V_{\text{LSB}} = 4.3$ mV at $V_{\text{DD}} = 1.1$ V) every ≈ 4 s — manageable with ≤ 1 s SRAM refresh; at 7 nm ($t_{\text{ox,eq}} \approx 1.5$ nm) tunnelling is $\approx 10^4 \times$ higher: a MOS gate cap loses one LSB in under 1 ms.

The 28 nm solution is BEOL *metal-insulator-metal* (MIM) capacitance: ALD-deposited HfO_2 exhibits 10^{-4} – $10^{-5} \times$ MOS tunnelling at the same field. TSMC 28 nm MIM density is ≈ 6 fF/ μm^2 ; a 50 fF cap occupies $\approx 8 \mu\text{m}^2$, compatible with the $\approx 17 \mu\text{m}^2$ cell. At 7 nm, HfO_2 -based MIM density is ≈ 10 fF/ μm^2 ; a 20 fF element fits within the $\approx 5 \mu\text{m}^2$ cell.

A second constraint arises from Hebbian update resolution:

$$\Delta V_w^{\min} = \frac{V_{\text{DD}}}{256} = \frac{I_{\text{hebb}} t_{\text{pulse}}}{C_w}. \quad (5)$$

At 28 nm, a 50 fF MIM cap changes by ~ 28 mV per standard pulse (≈ 7 LSBs), acceptable because the Hebbian rule converges over many updates; coarser per-pulse steps extend the convergence window but do not prevent convergence.

For non-volatile weight retention at and below 7 nm, Table XI compares available technologies.

FeFET (HfO_2 with ferroelectric HZO gate stack, e.g., GF 22FDX-FE) stores weight non-volatily in the gate ferroelectric polarisation state, replacing crystallographic retention (years) for the MOS tunnelling floor. PCM ($\text{Ge}_2\text{Sb}_2\text{Te}_5$) achieves comparable precision and endurance with more controllable intermediate resistance states, at the cost of higher write energy ($\sim$ pJ vs. $\sim$ fJ for FeFET). MRAM provides unlimited write endurance but is inherently binary; 8-bit reso-

TABLE XII: Inference energy at equivalent parameter scale (~ 340 M weights; architecturally comparable, not task-equivalent). H100 energy is HBM-bandwidth-dominated at batch = 1. PCN values are pre-silicon projections.

System	Energy/inference	Latency	On-chip learning
H100 SXM5 (batch = 1)	~ 6.8 J	$\sim 200 \mu\text{s}$	No
H100 SXM5 (batch = 128)	~ 53 mJ/sample	—	No
PCN 28 nm $\times 6$ chips	$\sim 7.7 \mu\text{J}$	$\sim 150 \mu\text{s}$	Yes
PCN 7 nm $\times 2$ chips	$\sim 3 \mu\text{J}$	$\sim 100 \mu\text{s}$	Yes

lution requires eight cells plus a local DAC. eFlash endurance (10^3 – 10^5 cycles) is incompatible with continuous Hebbian updating.

The fundamental challenge for NVM in an online learning context is write frequency: at 5,000 inferences/second with per-step weight updates, even 10^6 -cycle endurance is exhausted in minutes. A *hybrid architecture* sidesteps this: MIM capacitor and SRAM handle high-frequency Hebbian updates; NVM commits occur periodically (every $\sim 1,000$ steps, reducing NVM write rate to $\sim 5/\text{s}$). At 10^8 -cycle endurance, 5 writes/s gives a 5.5-year device lifetime. The PCN’s self-correcting Hebbian rule provides additional resilience: even if NVM restores weights at 4–5 bit precision, the learning rule re-converges to the optimal state within a small number of inference steps.

E. Energy Advantage Over Digital Inference

The zero off-chip weight bandwidth property creates a structural energy advantage for small-batch inference, where GPU operation is memory-bandwidth-limited.

An H100 GPU must read all model weights from HBM for every inference pass. For a 340-million-parameter model (BERT-large scale) at FP16: $340 \text{ M} \times 2 \text{ bytes} \times 10 \text{ pJ/byte} \approx 6.8 \text{ J}$ per forward pass at batch = 1 [2]. Amortised over batch = 128 the HBM cost falls to $\approx 53 \text{ mJ}$ per sample. An equivalent PCN deployment (six 28 nm chips, $6 \times 1.6 \text{ M}$ effective weights, $N_{\text{virt}} = 100$) operates at 51 mW with 150 μs inference time, consuming $\approx 7.7 \mu\text{J}$ per input with zero weight traffic.

At batch = 1 the energy ratio is approximately $10^6 \times$ in favour of the PCN, reflecting the GPU bandwidth wall rather than a raw compute advantage. At batch = 128 the HBM cost amortises and the ratio falls to $\sim 7,000 \times$; at very large batches, where GPU arithmetic units are compute-bound, the gap narrows further. The PCN advantage is most pronounced for real-time edge inference (batch = 1, latency-sensitive), continual on-device learning (weights update in place after every sample), and multi-chip deployment where weight synchronisation would otherwise dominate energy cost. The TOPS/W figures in Table X reflect the bandwidth-inclusive view: no energy is charged against a memory hierarchy. PCN at 7 nm reaches $\sim 3.8 \text{ TOPS/W}$ (same die), bracketing the H100’s 2.83 TOPS/W (INT8, compute-bound, large batch).

F. Scale by Area: A Complementary Scaling Strategy

Geometry scaling and area scaling are independent levers, and for an analog architecture they are not equally attractive. Geometry reduction shrinks cell area but simultaneously tightens VDD headroom, quantises transistor widths at FinFET nodes ($\leq 22 \text{ nm}$), and compounds the weight storage problems addressed in Section X-D. Area scaling — larger dies, multi-die modules, or wafer-scale integration — adds parameter capacity without touching the analog operating point.

The PCN architecture is unusually well-suited to area scaling for two reasons. First, the architecture is *locally connected*: MAC cells interact only with spatial neighbours and their own virtual layer stack. There is no global all-to-all computation and no backpropagation gradient broadcast. Adding cells requires no new long-range communication infrastructure. Second, the power density is very low: at 7 nm on a 25 mm^2 die, the analog core consumes $\approx 50 \text{ mW}$ ($\approx 2 \text{ mW/mm}^2$), $\sim 250 \times$ lower than the Cerebras WSE-3 ($\approx 500 \text{ mW/mm}^2$ [15]). A wafer-scale integration of the PCN tile at 7 nm on a 300 mm wafer ($\approx 57,000 \text{ mm}^2$ usable) yields ≈ 460 billion effective weights at $\approx 114 \text{ W}$ total. Crucially, the local Hebbian update rule requires no global gradient bus across the wafer, a key simplification over backprop-based wafer-scale designs.

A two-tier system follows naturally from the existing inter-chip protocol. *Edge nodes* at 28 nm or 130 nm operate at tens of milliwatts, learn local patterns from sensor inputs, and forward compressed representations upward. *Datacenter assemblies* — large MCM or wafer-scale systems — receive these representations and generate top-down predictions using a much larger weight capacity. The dynamic routing mechanism (Section VI) allows inter-tier connection topology to self-organise from data using the same Hebbian rule as intra-chip weights, replicating the structural principle of biological predictive coding hierarchies in scalable silicon.

The 28 nm planar node is the practical sweet spot for this strategy: the last geometry with continuous transistor width sizing, adequate VDD headroom (1.1 V supports the 5T OTA with margin), and MIM capacitors sufficient for SRAM-refresh-backed weight storage. Area scaling at 28 nm is a simpler, more manufacturable path to equivalent parameter counts than geometry reduction below 28 nm.

XI. ALGORITHMIC VALIDATION

We validate the chip’s learning dynamics using a Python/NumPy software model that faithfully mirrors the hardware’s computational graph (`sim/pcn_core.py`, `sim/circuit_sim.py`, `sim/pcn_predict.py`).

A. Circuit-Level Validation

Seven circuit-level experiments verify the MAC model against hardware simulation parameters:

Three design rules emerge from simulation failures:

Operating range. V_{row} must remain within $[0.4, 1.4] \text{ V} = V_{\text{cm}} \pm 0.5 \text{ V}$. For unit- ℓ_2 row weights a firmware scale factor ≈ 0.036 is applied during `inp_dac` reload.

TABLE XIII: Circuit-level validation (7 experiments, hardware-calibrated model; $G_0 = 6.82$ V/V, $V_{cm} = 0.9$ V, $V_{DD} = 1.8$ V, ADC/DAC range $[0.4, 1.4]$ V)

Exp	Description	Key result	Status
C1	Single-cell transfer curve	$G_0 = 6.82$ V/V	PASS
C2	16-cell KCL summation	Superposition holds ($R^2 = 1.00$)	PASS
C3	Spatial 4-layer cascade	mod0/1/2/3 = 1.43/0.45/0.70/0.54 V/V	PASS
C4	Temporal 4-VL cascade	$V_{in,cm} = 0.907\text{--}0.925$ V; 6.82 V/V	PASS
C5	Spatial vs. temporal	Temporal 6.68 V/V, reset-guaranteed	PASS
C6	ADC+DAC round-trip	Error = 0.50 LSB, SNR = 24.8 dB	PASS
C7	16-cell dot product	$R^2 = 1.000$, residual $\equiv 0$	PASS

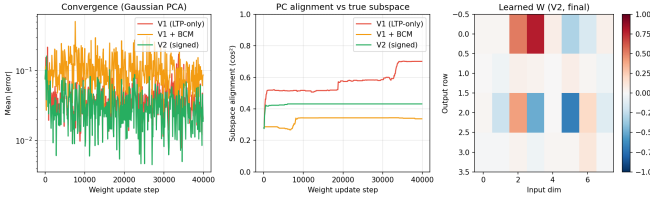


Fig. 1: E1: Mode comparison on a Gaussian PCA task ($n_{in} = 8$, $n_{rows} = 4$, SNR = 20 dB, 8 epochs). V1 (LTP-only) achieves 0.70 subspace alignment and converges to a gate-quiescent fixed point; V2 (signed, four-quadrant) reaches 0.43 alignment.

ADC/DAC reference co-alignment. Mismatched references produce > 44 LSB round-trip error; matched references give 0.50 LSB (C6 above).

Minimum learning rate. Below $LR_{min} \approx 0.01$ for 8-bit Hebbian encoding, convergence stalls; the hardware Hebbian step (≈ 5 mV/pulse) is calibrated to exceed this threshold.

B. Software Simulation — Mode Comparison

V1 (LTP-only) achieves final subspace alignment 0.70 and prediction error 0.021, converging to a gate-quiescent state. V2 (signed) achieves 0.43 alignment; discrete weight steps introduce oscillations near the fixed point when equal-magnitude LTP and LTD updates compete. V2 with competitive 2-winner-take-all gating achieves 1.00 template selectivity on a 16-dimensional dictionary-learning task. 8-bit weight quantisation reduces reconstruction MSE from 0.018 to 0.287 but preserves subspace alignment (0.41 vs. 0.43 float), confirming that 6.6-effective-bit weight resolution does not prevent principal subspace extraction.

C. Temporal Reuse Convergence

All four virtual layers converge; deeper layers receive progressively more residualised input and reach lower final error. $V_{in,cm}$ remains within ± 25 mV of V_{cm} across all layers, confirming the operating-point reset.

D. Multi-Cell Predictive Network Validation

A two-layer undercomplete PCN ($\mathbf{W}_0: 8 \times 16$, $\mathbf{W}_1: 8 \times 8$) is trained with GHA on Layer 0 and $k = 1$ WTA Oja on

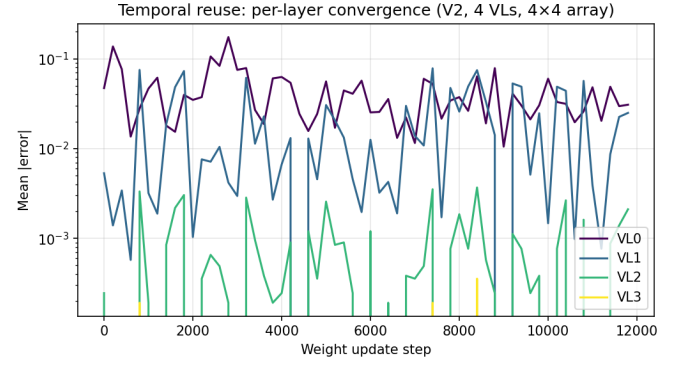


Fig. 2: E4: Temporal reuse convergence (4-virtual-layer stack, 4×4 array, V2 mode, 3 epochs $\times$ 4,000 samples). Per-layer prediction errors decrease monotonically: VL0=0.031, VL1=0.025, VL2=0.002, VL3 ≈ 0 .

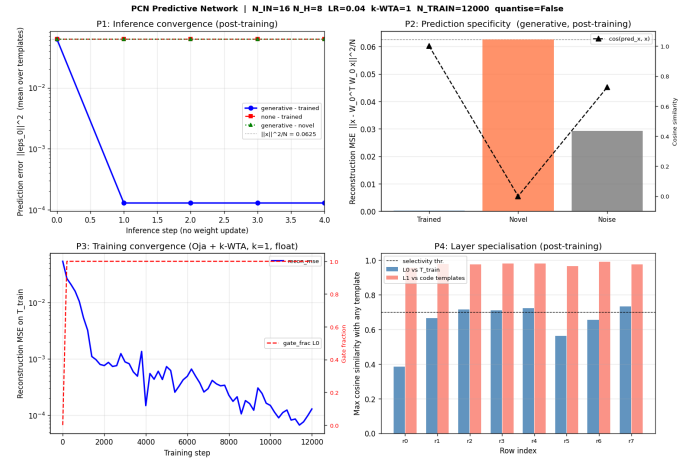


Fig. 3: Multi-cell predictive network ($N_{in} = 16$, $N_H = 8$, GHA + cosine-decaying LR, 12,000 steps). P1: single-step inference error ratio 484 $\times$. P2: trained MSE 483 $\times$ below novel. P3: training convergence. P4: Layer-1 100% row selectivity in code space.

Layer 1. Eight orthonormal templates and eight orthogonal novel templates are generated by QR decomposition.

Results. After training, a single generative inference step reduces prediction error from $0.0625 = \|\mathbf{x}\|^2/N$ to 1.3×10^{-4} — a 484 $\times$ reduction. Novel templates maintain prediction error 0.0625 at all inference steps: the network makes no prediction for inputs it has not seen. Trained-template MSE = 1.3×10^{-4} vs. novel MSE = 0.0625 gives 483 $\times$ specificity. Layer 1: 100% row selectivity ($\cos > 0.7$); Layer 0: 99.8% subspace coverage.

Correct training rule. Three simulation failures during development identified the correct rule. (i) Without deflation, all rows converge to the same principal direction. (ii) k -WTA alone creates dead neurons: rows receiving negative Oja updates cannot recover. (iii) Chaining $\mathbf{W}_1^T \mathbf{W}_1$ in the prediction path amplifies by up to $N_H \times$, causing inference divergence. The correct architecture uses GHA on Layer 0 (guaranteed orthonormal convergence), excludes Layer 1 from

TABLE XIV: Comparison to prior neuromorphic hardware

Work	Tech	Learning	Capacity	Architecture
Loihi 2 [11]	7 nm	STDP	digital SRAM	spike-based
BrainScaleS-2 [12]	65 nm	backprop	64 syn/mm ²	mixed-signal
DynapSEL [16]	180 nm	STDP	~40 syn/mm ²	current-mode seq.
Oh [1]	Digital RTL	Supervised PC	SRAM weights	MAC+FSM
This work	130 nm	Hebbian PCN	256 × N_v /cell	temporal + routing

the prediction path ($\hat{\mathbf{x}} = \mathbf{W}_0^\top \mathbf{W}_0 \mathbf{x}$, a bounded projector), and applies k -WTA only to Layer 1. All six quantitative checks pass.

XII. DISCUSSION

A. Five-Transistor Topology and Its Consequences

The five-transistor OTA couples tail current to weight voltage and input common-mode through MN3. This causes gain variation across spatial layers in the untrained state (mod0 = 1.43, mod1–3 = 0.45–0.70 V/V) and couples power draw directly to weight encoding. Both are self-corrected by Hebbian learning; temporal reuse additionally eliminates cascade-depth drift by resetting V_{cm} at each virtual layer. A Gilbert-cell topology [13] with independent tail biasing would eliminate both couplings structurally in a next-generation cell.

B. Practical Scope of the Proof-of-Concept

The Sky130A chip implements one virtual tile: a single 16×16 physical array cycling through up to 8 virtual layers with SRAM-backed weight persistence, Hebbian learning, and a Wishbone CPU interface. It is functionally complete as a feature extractor or first-layer encoder. A full autonomous PCN additionally requires the upward transpose product $\mathbf{W}^\top \varepsilon$ for representation update, which would require a second MAC array or time-multiplexed transpose access — a substantial extension left for future work.

C. Dynamic Routing: Simulation Status

The routing Hebbian write mechanism is verified in SPICE (Section VI). End-to-end convergence of routing weights to a functionally correct learned topology requires a testbench that simultaneously trains MAC weights and routing weights over many inference cycles. The theoretical basis — routing weights as Hebbian MAC weights at a different hierarchical level — is well-founded on the existing simulation evidence, but experimental demonstration is deferred to future work.

D. Comparison to Prior Neuromorphic Hardware

The structural differences from prior work are: (i) rate-coded rather than spike-coded learning, enabling continuous activations and the PCN hierarchical generative model; (ii) temporal multiplexing, decoupling weight capacity from analog area; (iii) dynamic routing at the inter-module level; (iv) zero off-chip weight bandwidth at inference, regardless of effective network depth.

E. Claims, Evidence, and Open Questions

The evidence base spans three levels, and distinguishing them is important for assessing the work’s current standing.

SPICE-verified (circuit level). Seven circuit checks (Table XIII) validate individual block behaviour against analytical predictions using Sky130A PDK models. The PMOS source-follower level shift (+0.67 V), spatial cascade gains (mod0–3), and temporal reuse gain (6.68 V/V) are direct SPICE measurements. All results are reproducible from the published netlist.

Tool-verified (digital synthesis and P&R). 1,117 standard cells, +8.46 ns setup margin, zero routing DRC violations. These are tool outputs, independent of simulation assumptions, and reproducible from the published RTL.

Software-verified (algorithmic simulation). 484× prediction error reduction and 483× specificity are outputs of open-source Python code on a hardware-calibrated model. These results are established on synthetic Gaussian tasks; generalisation to practical data distributions is an open question that hardware validation will address.

Pre-silicon projections (estimated). All absolute power figures, node-scaling projections (Tables X, IX), TOPS/W comparisons, and NVM technology projections in Section X-D are engineering estimates from standard scaling rules and published literature values. They have not been validated by analog simulation at smaller geometries. The energy comparison (Table XII) correctly reflects the memory-bandwidth argument but compares architecturally different systems at different numerical precision and task scope; it is not a task-equivalent benchmark.

Path to silicon validation. Tape-out via Efabless OpenMPW is targeted as the primary validation step. The critical experiment is end-to-end multi-cell operation: a full 16×16 array with SRAM-backed temporal reuse and live Hebbian updates, measured in silicon against the SPICE-calibrated software predictions. All design files, RTL, and simulation scripts are open-source, enabling independent verification of every circuit and algorithmic claim prior to fabrication.

XIII. CONCLUSION

We have presented an analog VLSI architecture for Predictive Coding Networks that addresses the three key barriers to scalable analog neural computation: weight density (via temporal layer reuse), fixed topology (via learned routing weights), and quiescent power (via precision-gate co-scaling).

The Sky130A proof-of-concept is fully verified at the simulation level: SPICE-validated analog blocks, a 22-state digital controller synthesised in 1,117 standard cells, place-and-route closed at +8.46 ns setup margin with zero routing DRC violations. Software simulation with the hardware-calibrated model achieves 484× single-step prediction error reduction and 483× specificity with GHA-trained weights, and identifies the correct training rule and three hardware design rules from simulation failures.

The architecture scales naturally along two orthogonal dimensions. Temporal reuse decouples effective weight capacity from analog area; dynamic routing allows the network to

discover inter-chip prediction topology from data. Multi-chip systems scale linearly — chips are independent, the inter-chip interface is 8-bit digital SPI, and no synchronisation overhead arises as the system grows.

A second and independent scaling dimension is area rather than geometry. The PCN’s locally connected Hebbian update requires no cross-chip gradient synchronisation, so parameter capacity grows linearly with silicon area at a fixed node. At 28 nm — the last geometry with continuous transistor width sizing and adequate VDD headroom for the 5T OTA — a 25 mm² die holds ≈ 59 M effective weights at ≈ 50 mW. At wafer scale, the same tile pattern yields ≈ 460 billion effective weights at ≈ 114 W. The locality of Hebbian learning eliminates the global gradient-bus complexity that constrains backprop-based wafer-scale designs, positioning the PCN as a candidate architecture for large-scale analog learning systems where the scaling bottleneck is area rather than geometry.

All performance claims are pre-silicon simulation results; the Sky130A tape-out will provide the first silicon-validated evidence for the operating-point, gain, power, and learning dynamics described here.

REFERENCES

- [1] T. Oh, “A synthesizable RTL implementation of predictive coding networks,” *arXiv preprint arXiv:2603.18066*, 2026.
- [2] NVIDIA Corporation, “H100 SXM datasheet,” NVIDIA, Tech. Rep., 2022, tensor Core GPU, 700 W TDP, 3.35 TB/s HBM3 bandwidth.
- [3] R. P. Rao and D. H. Ballard, “Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects,” *Nature neuroscience*, vol. 2, no. 1, pp. 79–87, 1999.
- [4] R. Bogacz, “A tutorial on the free-energy framework for modelling perception and learning,” *Journal of mathematical psychology*, vol. 76, pp. 198–211, 2017.
- [5] C. Mead, *Analog VLSI and neural systems*. Addison-Wesley, 1989.
- [6] T. D. Sanger, “Optimal unsupervised learning in a single-layer linear feedforward neural network,” *Neural Networks*, vol. 2, no. 6, pp. 459–473, 1989.
- [7] T. Shibata and T. Ohmi, “A functional MOS transistor featuring gate-level weighted sum and threshold operations,” *IEEE Transactions on Electron Devices*, vol. 39, no. 6, pp. 1444–1455, 1992.
- [8] C. Diorio, P. Hasler, B. A. Minch, and C. A. Mead, “A single-transistor silicon synapse,” *IEEE Transactions on Electron Devices*, vol. 43, no. 11, pp. 1972–1980, 1996.
- [9] S. Ambrogio, P. Narayanan, H. Tsai, R. M. Shelby, I. Patterson, S. Lin, R. Carboni, S. Nandakumar, T. Nishi, A. Woicik *et al.*, “Equivalent-accuracy accelerated neural-network training using analogue memory,” *Nature*, vol. 558, no. 7708, pp. 60–67, 2018.
- [10] M. Prezioso, F. Merrih-Bayat, B. Hoskins, G. Adam, K. Likharev, and D. Strukov, “Training and operation of an integrated neuromorphic network based on metal-oxide memristors,” *Nature*, vol. 521, no. 7553, pp. 61–64, 2015.
- [11] M. Davies, N. Srinivasa, T.-H. Lin, G. Chinya, Y. Cao, S. H. Choday, G. Dimou, P. Joshi, N. Lines, R. Mathaikutty *et al.*, “Loihi: A neuromorphic manycore processor with on-chip learning,” *IEEE Micro*, vol. 38, no. 1, pp. 82–99, 2018.
- [12] M. Pfeiffer and T. Pfeil, “Deep learning with spiking neurons: Opportunities and challenges,” *Frontiers in neuroscience*, vol. 12, p. 774, 2018.
- [13] B. Gilbert, “A precise four-quadrant multiplier with subnanosecond response,” *IEEE Journal of Solid-State Circuits*, vol. 3, no. 4, pp. 365–373, 1968.
- [14] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [15] Cerebras Systems, “Cerebras WSE-3: The third generation wafer-scale engine,” Cerebras Systems, Tech. Rep., 2023, 900,000 cores, 44 GB on-chip SRAM, 23 kW system power, 46,225 mm² die.
- [16] E. Chicca, F. Stefanini, C. Bartolozzi, and G. Indiveri, “Neuromorphic electronic circuits for building autonomous cognitive systems,” *Proceedings of the IEEE*, vol. 102, no. 9, pp. 1367–1388, 2014.